

OSANDA ABHIRU ILLEPERUMA

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EDUCATION

Reading PhD in Mathematics, Indiana University, Bloomington, USA *August 2022 - Present*
CGPA: 3.89/4.00

MA in Mathematics, Indiana University, Bloomington, USA *August 2022 - February 2024*
CGPA: 3.75/4.00

BSc Honors Degree in Mathematics, University of Colombo, Sri Lanka *January 2017 - May 2021*
CGPA: 3.94/4.30

SKILLS

- **Problem Solving**
- **Mathematical Communication**
- **Programming skills:** Python, Java
- **Theorem Provers:** Lean (Beginner)
- **Github**

RESEARCH EXPERIENCE

Category Theoretic Applications in Computer Science

Advisor : Dr. Jayampathy Ratnayake

- The research focused on surveying the automata theory in computer science within a category-theoretic framework.

Model Theoretic Algebra, Differential Algebra, Initial and Final Coalgebras(PhD research project currently working on)

Advisor : Dr. Lawrence S. Moss

- The research focuses on answering a question related to coalgebra of streams using algebraic methods.

PUBLICATIONS

- **The Universal Property of $R[[X]]$ via Corecursive Algebra**, in preparation

WORKING EXPERIENCE

Associate Instructor, Department of Mathematics, Indiana University *May 2023 - Present*

Graduate Teaching Assistant, Department of Mathematics, Indiana University *January 2023 - May 2023*

Temporary Assistant Lecturer, Department of Mathematics, University of Colombo *November 2021 - May 2022*

Teaching Assistant, Department of Mathematics, University of Colombo *May 2021 - November 2021*

ACADEMIC ACHIEVEMENTS

- Erdős Institute Participation Scholarship - 2026
- Glenn Schober Memorial Fellowship - 2026
- Hazel King Thompson Fellowship - 2023
- College of Arts and Sciences Fellowship - 2022
- Dharmadasa Punchihewa Memorial Prize for Mathematics - 2020
- Gulamhusein A. J. Noorbhai Gold Medal for the Best Student in Mathematics - 2020
- Douglas Amarasekara Prize for the Best Student in Mathematics - 2020
- Mr. and Mrs. D. P. Epasinghe Memorial Gold Medal for Mathematics - 2020
- Professor Valentine Joseph Memorial Gold Medal for the Best Student in Mathematics - 2020

EXTRA-CURRICULAR ACTIVITIES

Volunteering:

Local Organization Committee Member, CALCO 2023 & MFPS XXXIX.

2023

- Member of the local organization committee of the 10th Conference on Algebra and Coalgebra in Computer Science (CALCO 2023) and the 39th Conference on the Mathematical Foundations of Programming Semantics (MFPS XXXIX) took place together at Indiana University Bloomington on June 19 – 23, 2023.

President - Epsilon Delta Society, Department of Mathematics, University of Colombo.

2019-2020

- Organizing TIMO and HKIMO Heats Award Ceremony-2019 in collaboration with the Sri Lanka Olympiad Mathematics Foundation
- Organizing a series of workshops for Sri Lankan school students aiming the Sri Lanka Mathematics Competition(SLMC), Thailand International Mathematical Olympiad(TIMO), Hong Kong International Mathematical Olympiad(HKIMO) and Hong Kong International Science Olympiad(HKISO) in collaboration with the Sri Lanka Olympiad Mathematics Foundation in 2019.
- Initiating the problem-solving session “A Date with Putnam” for Faculty of Science undergraduates.

President - Interact Club, St. Aloysius' College-Galle.

2012-2013

Member, Gavel Club, .

2018-2021

Sports and Hobbies:

Chess - Championship at Inter Faculty Freshers' Tournament.

Badminton

Music - Playing Violin, Guitar and Keyboard.

I hereby attest that the above details are true and accurate to the best of my knowledge .

Osanda Abhiru Illeperuma